



GRADE 8
Computational Thinking
National Level Examination (NLE) 2026

SUBJECT CODE

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Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And Hb Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Abstract & Spatial Thinking (10 Questions); Section B: Pattern Recognition (10 Questions); Section C: Decomposition, Data & Constraints (10 Questions); Section D: Algorithmic Thinking & Logical Flow (10 Questions); Section E: Ai Literacy & Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. What is the decimal value of the binary number 101101?

- A. 41
- B. 43
- C. 45
- D. 47

Q2. A number N leaves remainder 4 when divided by 7. What remainder will $3N + 2$ leave when divided by 7?

- A. 0
- B. 2
- C. 4
- D. 6

Q3. A $3 \times 3 \times 3$ cube is painted on all outer faces and cut into unit cubes. How many unit cubes have exactly two painted faces?

- A. 8
- B. 12
- C. 16
- D. 20

Q4. The point $(3, -2)$ is rotated 90° counter-clockwise about the origin. What are its new coordinates?

- A. $(-2, -3)$
- B. $(2, 3)$
- C. $(-2, 3)$
- D. $(2, -3)$

Q5. Reflecting a figure in a vertical line and then in a horizontal line through the same centre is equivalent to a rotation of:

- A. 90°
- B. 120°
- C. 180°
- D. 270°

Q6. A solid rectangular block is made of $3 \times 2 \times 2$ unit cubes. How many unit-square faces are exposed on its surface?

- A. 24
- B. 28
- C. 32
- D. 36

Q7. A regular octagon is rotated about its centre. What is the smallest positive angle that leaves it unchanged?

- A. 30°
- B. 45°
- C. 60°
- D. 90°

Q8. A rectangular grid has 4 rows and 3 columns of small rectangles. How many grid intersection points are there?

- A. 12
- B. 16
- C. 20
- D. 24

Q9. Which of the following Capital letter has both horizontal vertical line of symmetry.

- A. R
- B. P
- C. H
- D. F

Q10. The point $(5, -3)$ is reflected across the y-axis. What are its new coordinates?

- A. $(-5, -3)$
- B. $(5, 3)$
- C. $(-3, 5)$
- D. $(3, -5)$

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 2, 5, 12, 27, 58, ____.

- A. 115
- B. 119
- C. 121
- D. 123

Q12. Find the next number: 1, 2, 6, 15, 31, 56, ____.

- A. 81
- B. 86
- C. 92
- D. 96

Q13. These are binary numbers in order: 1, 10, 11, 100, 101, 110, 111, _____. What comes next?

- A. 1000
- B. 1001
- C. 1010
- D. 1111

Q14. Find the next number: 7, 14, 12, 24, 22, 44, 42, ____.

- A. 62
- B. 82
- C. 84
- D. 86

Q15. A rule maps n to $n^2 + n$. What is the output when $n = 8$?

- A. 64
- B. 70
- C. 72
- D. 80

Q16. The letters B, C, E, G, K, M, Q are at prime-numbered positions in the alphabet. Which letter comes next?

- A. R
- B. S
- C. T
- D. U

Q17. Find the next number: 128, 64, 32, 16, 8, ____.

- A. 2
- B. 4
- C. 6
- D. 12

Q18. Find the next number: 3, 7, 15, 31, ____.

- A. 47
- B. 55
- C. 63
- D. 71

Q20. Find the next pair: B2, E6, I12, N20, ____.

- A. S28
- B. T30
- C. U30
- D. U32

Q19. Find the next number: 2, 6, 12, 20, 30, 42, ____.

- A. 54
- B. 56
- C. 58
- D. 60

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q21. What is the smallest number greater than 20 that leaves remainder 2 when divided by 5 and remainder 1 when divided by 3?

- A. 21
- B. 22
- C. 27
- D. 32

Q26. A dataset of 1200 records is split 70% for training, 15% for validation and the rest for testing. How many records are used for testing?

- A. 150
- B. 180
- C. 200
- D. 240

Q22. Five identical machines make 300 items in 4 hours. At the same rate, how many items will 3 machines make in 6 hours?

- A. 240
- B. 270
- C. 300
- D. 360

Q27. In a class of 50 students, 28 like Coding, 24 like Robotics and 12 like both. How many like neither?

- A. 8
- B. 10
- C. 12
- D. 14

Q23. The mean of 12, 18, 20, 25 and x is 20. What is x?

- A. 20
- B. 23
- C. 25
- D. 27

Q28. The numbers 4, 7, 9, 12, 15, 18, 21 have what median?

- A. 9
- B. 12
- C. 15
- D. 18

Q24. Two boxes A and B contain balls in the ratio 5 : 3, with 64 balls altogether. If 8 balls are moved from A to B, what is the new ratio A : B?

- A. 1 : 1
- B. 3 : 2
- C. 4 : 3
- D. 5 : 4

Q29. A dataset contains 800 rows. If 65% are training and 20% are validation, how many are test rows?

- A. 100
- B. 120
- C. 140
- D. 160

Q25. An item marked ₹1600 gets a 15% discount. Then 5% tax is added to the discounted price. What is the final price?

- A. ₹1360
- B. ₹1400
- C. ₹1428
- D. ₹1440

Q30. A 128×64 grayscale image uses 8 bits per pixel. Ignoring metadata, how many bytes are needed?

- A. 4096
- B. 8192
- C. 16384
- D. 65536

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q31. Apply this rule four times: if divisible by 3, divide by 3; otherwise if even, divide by 2; otherwise add 5. Start with 28. What is the result?

- A. 2
- B. 4
- C. 6
- D. 8

Q33. Set $S = 0$. For $n = 1$ to 5: if n is even, add $2n$ to S ; otherwise add n . What is the final value of S ?

- A. 18
- B. 20
- C. 21
- D. 24

Q32. Start with $x = 3$. Repeat four times: $x = 2x + 1$. What is the final value of x ?

- A. 31
- B. 47
- C. 63
- D. 65

Q34. On a grid, a robot goes from (1,1) to (4,4) using only Right and Up moves. It may not pass through (2,2). How many shortest paths are possible?

- A. 6
- B. 8
- C. 10
- D. 12

B. 28

Q35. Four tasks obey: B is before A; A is before C; C is before D. Which task must be third?

- A. A
- B. B
- C. C
- D. D

Q38. Start $x = 20$. Repeat: $x = x/2$ while x is even. What value stops the loop?

- A. 2
- B. 4
- C. 5
- D. 10

Q36. To find the largest of a , b and c , an algorithm sets $\text{max}=a$, then uses 'if $b < \text{max}$, $\text{max}=b$ '. What should replace ' $<$ ' to fix it?

- A. =
- B. >
- C. <=
- D. !=

Q39. A shortest path needs exactly 4 Right moves and 2 Up moves. How many different shortest paths are possible?

- A. 10
- B. 12
- C. 15
- D. 18

Q37. Set count = 0. For n from 1 to 10, increase count when n is divisible by 3. What is count at the end?

- A. 2
- B. 3
- C. 4
- D. 5

Q40. An algorithm adds numbers 1 to 5 but accidentally resets $\text{sum}=0$ inside every loop iteration. What is the final displayed sum?

- A. 0
- B. 5
- C. 10
- D. 15

SECTION E – OLYMPIAD CHALLENGE

Q41. Which is the correct conceptual order of an AI project lifecycle?

- A. Collect → Define → Reflect → Test
- B. Define → Collect → Test → Reflect/Improve
- C. Test → Collect → Define → Stop
- D. Define → Test → Collect → Stop

Q46. What is $11101_2 + 1011_2$ in binary?

- A. 100100_2
- B. 101000_2
- C. 101100_2
- D. 110000_2

Q42. An AI labels emails as 'spam' or 'not spam'. Which technique is this?

- A. Classification
- B. Regression
- C. Clustering
- D. Compression

Q47. Which AI domain is mainly used to identify objects in photographs?

- A. NLP
- B. Computer Vision
- C. Audio synthesis
- D. Spreadsheet formulas

Q43. An AI predicts the selling price of a house from its size, location and age. Which technique is most suitable?

- A. Classification
- B. Regression
- C. Clustering
- D. Translation

Q48. A game-playing agent improves by receiving rewards for good moves. This is:

- A. Supervised learning
- B. Reinforcement learning
- C. Clustering
- D. Compression

Q44. A system groups unlabeled news articles by similarity of topic. Which method best describes this?

- A. Classification
- B. Regression
- C. Clustering
- D. Encryption

Q49. Why is a validation dataset useful?

- A. To replace all training data
- B. To tune/check the model before final testing
- C. To hide labels permanently
- D. To increase screen resolution

Q45. A face-recognition dataset contains very few examples from one age group. What is the most important risk?

- A. Larger files
- B. Unfair or lower accuracy for that group
- C. Slower internet
- D. Inability to store numbers

Q50. Which is the safest data practice?

- A. Collect all possible personal data
- B. Share raw student data publicly
- C. Collect only necessary data with consent
- D. Store passwords in plain text